Forward Collision Warning – High-Level Design

AUTOSAR Adaptive/Classic Interface | Sample Reference Architecture Document

Version 1.0 | September 2026
Prepared for internal engineering review and pilot evaluation

AUTOSAR Classic Platform | High-Level Design Document

Document Control

Field	Details
Document Title	Forward Collision Warning – High-Level Design
Version	1.0
Document Type	Software Architecture / High-Level Design (HLD) Document
Purpose	Describe the software architecture, sensor interfaces, signal flow, and safety-relevant dependencies of the Forward Collision Warning (FCW) feature.
Intended Audience	System architects, functional safety engineers, software developers, test engineers
Status	Sample / Reference (Synthetic Data)
Confidentiality	Internal use (sample/synthetic data for pilot evaluation)

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Introduction and Scope

This document describes the High-Level Design of the Forward Collision Warning (FCW) feature, part of the ADAS domain controller software. FCW uses front radar and camera fusion data to estimate time-to-collision and issue driver warnings. This is a safety-relevant function (target ASIL B) and interfaces with the braking and HMI domains.

Scope covers the following software components:

- SWC_SensorFusion
- SWC_CollisionRiskEstimator
- SWC_WarningArbitration
- SWC_HMIAlertOutput

Software Component Overview

Component Name	Responsibility	Mapped ECU
SWC_SensorFusion	Fuses radar and camera object lists into a unified track list	ADAS_Domain Ctrl_01
SWC_CollisionRiskEstimator	Computes time-to-collision (TTC) and risk level	ADAS_Domain Ctrl_01
SWC_WarningArbitration	Arbitrates warning level against driver-override state	ADAS_Domain Ctrl_01
SWC_HMIAlertOutput	Drives audible and visual driver warning outputs	ADAS_Domain Ctrl_01

Ports and Interfaces

Port Name	Direction	Interface Type	Owning Component
RPort_RadarObjectList	Required	Sender-Receiver	SWC_SensorFusion
RPort_CameraObjectList	Required	Sender-Receiver	SWC_SensorFusion
PPort_FusedTrackList	Provided	Sender-Receiver	SWC_SensorFusion
RPort_FusedTrackList	Required	Sender-Receiver	SWC_CollisionRiskEstimator
PPort_TTCEstimate	Provided	Sender-Receiver	SWC_CollisionRiskEstimator
RPort_DriverOverrideState	Required	Sender-Receiver	SWC_WarningArbitration
PPort_WarningLevel	Provided	Sender-Receiver	SWC_WarningArbitration
CSPort_HMIDisplayService	Required	Client-Server	SWC_HMIAlertOutput

Signals and Data Elements

Signal Name	Data Type	Range / Values	Description
FusedTrackList	struct[16]	Up to 16 tracked objects	Fused radar+camera object array
TTCEstimate	float32	0.0–10.0 s	Estimated time-to-collision
RiskLevel	uint8	0 = None, 1 = Low, 2 = Medium, 3 = High	Computed collision risk level
DriverOverrideState	boolean	TRUE / FALSE	Driver braking/steering override flag
WarningLevel	uint8	0–3	Arbitrated warning level to HMI
HMIAlertType	uint8	0 = None, 1 = Visual, 2 = Audible, 3 = Both	Output alert modality

Component Dependencies

Component	Depends On	Dependency Type
SWC_SensorFusion	BSW_CAN, BSW_Ethernet (camera link)	Raw sensor data reception
SWC_CollisionRiskEstimator	SWC_SensorFusion	Consumes fused track list
SWC_WarningArbitration	SWC_CollisionRiskEstimator, BSW_VehicleSpeedSignal	Risk input and vehicle state
SWC_HMIAlertOutput	SWC_WarningArbitration, BSW_DiagnosticEventManager	Warning level input, fault reporting

Functional Flow – Warning Generation Sequence

- 1. Radar and camera object lists are received over CAN/Ethernet by SWC_SensorFusion.
- 2. SWC_SensorFusion produces a unified FusedTrackList at a fixed 20 ms cycle.
- 3. SWC_CollisionRiskEstimator computes TTCEstimate for the most relevant target.
- 4. RiskLevel is derived from TTCEstimate against calibrated thresholds.
- 5. SWC_WarningArbitration checks DriverOverrideState before elevating WarningLevel.
- 6. SWC_HMIAlertOutput maps WarningLevel to an HMIAlertType and triggers the display/audio service.

Safety-Relevant Notes

- This function is preliminarily allocated ASIL B; final ASIL confirmed via HARA (out of scope for this document).
- TTC threshold calibration is placeholder data and must be validated through track testing.
- Freedom-from-interference between SWC_WarningArbitration and non-safety HMI components is assumed but not yet verified.

Revision History

Version	Date	Description
0.1	August 2026	Initial component breakdown
1.0	September 2026	Added interfaces, signals, dependencies, and safety notes